7. VLAN CONFIGURATION, SWITCH CONFIGURATION AND INTER-VLAN ROUTING

1. Objective

To configure VLANs on a switch, assign ports to VLANs, configure switchport modes (access and trunk), and implement Inter-VLAN routing using a router-on-a-stick configuration in Cisco Packet Tracer.

2. Topology Overview

The network consists of one router connected to a switch via a trunk link. The switch has four PCs connected to different ports, which are assigned to two VLANs (VLAN 10 and VLAN 20). Inter-VLAN routing is configured on the router using subinterfaces.

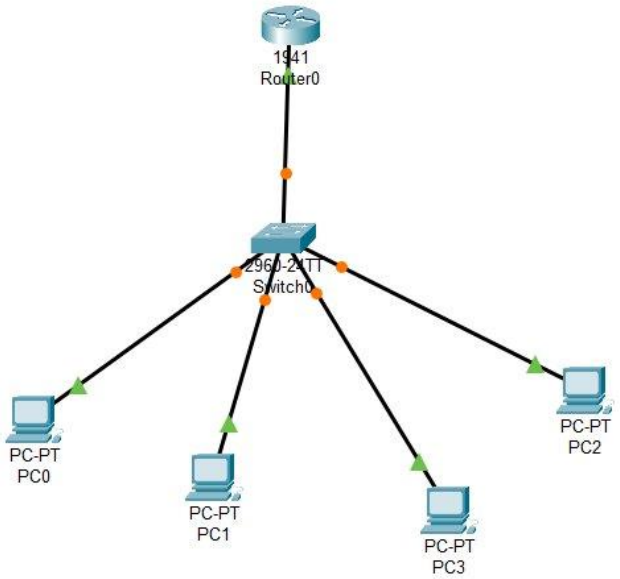

Figure 1: Network Topology Diagram

3. Devices Used

- 1 × Router (1941)
- 1 × Switch (2960)
- 4 × PC-PT (PC0, PC1, PC2, PC3)

4. VLAN and IP Addressing Scheme

The network is segmented into two VLANs:

VLAN ID	VLAN Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

4.1 Device IP Addressing Table

Device	Interface/Port	IP Address	VLAN
Router0	Gi0/0.10	192.168.10.1	10
Router0	Gi0/0.20	192.168.20.1	20
PC0	Fa0/1 (Switch)	192.168.10.2	10
PC1	Fa0/2 (Switch)	192.168.10.3	10
PC2	Fa0/3 (Switch)	192.168.20.2	20
PC3	Fa0/4 (Switch)	192.168.20.3	20

5. Configuration Steps

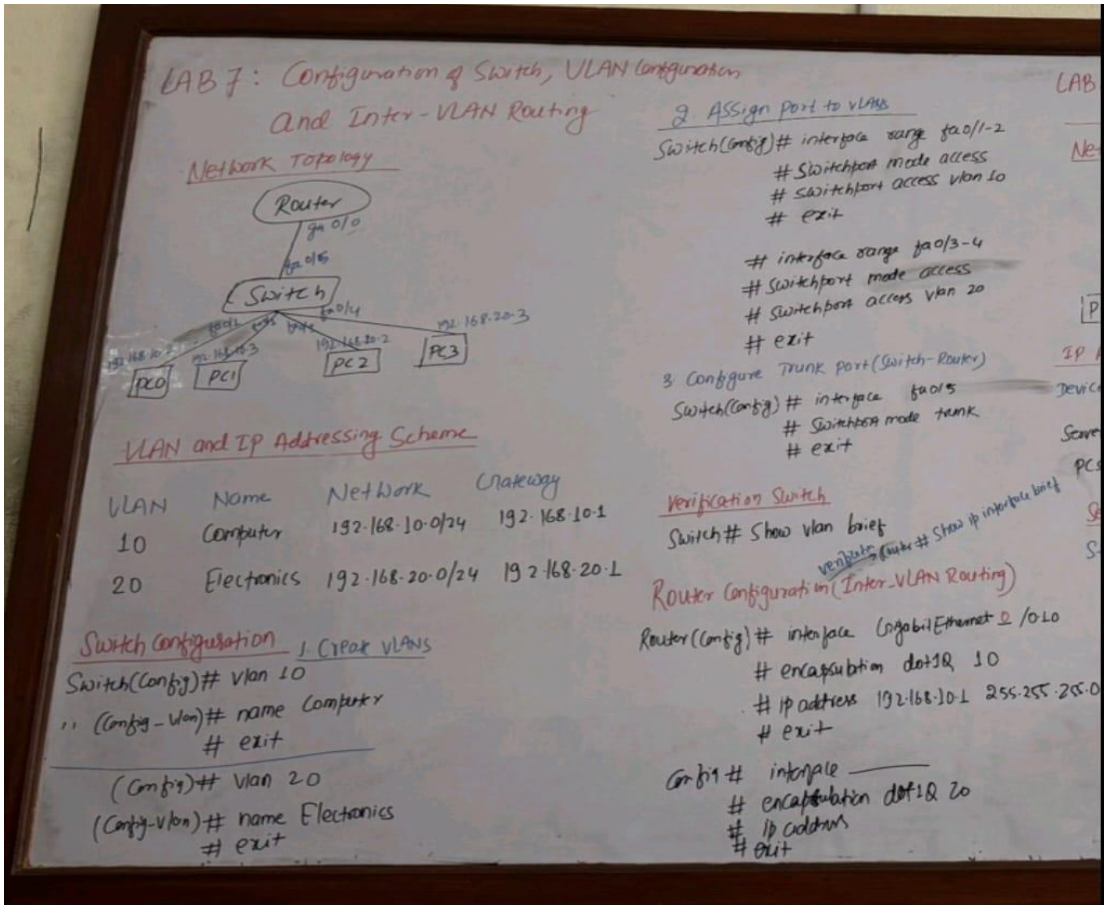

Figure 2: Configuration Notes

5.1 Switch VLAN Configuration

5.1.1 Create VLANs

Switch(config)# vlan 10

Switch(config-vlan)# name Computer

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name Electronics

Switch(config-vlan)# exit

5.1.2 Assign Ports to VLANs

For VLAN 10 (Ports Fa0/1-2):

Switch(config)# interface range fa0/1-2

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config-if-range)# exit

For VLAN 20 (Ports Fa0/3-4):

Switch(config)# interface range fa0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit

5.1.3 Configure Trunk Port

Switch(config)# interface fa0/5
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit

5.2 Router Inter-VLAN Configuration

5.2.1 Configure Subinterface for VLAN 10

Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

5.2.2 Configure Subinterface for VLAN 20

Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit

5.2.3 Enable Physical Interface

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit

5.3 PC IP Configuration

Configure each PC with static IP addresses as shown in the Device IP Addressing Table (Section 4.1), using their respective VLAN gateway addresses.

Subnet mask for all devices: 255.255.255.0

6. Verification Commands

6.1 Switch Verification

Switch# show vlan brief

Purpose: Displays VLAN configuration and port assignments

6.2 Router Verification

Router# show ip interface brief

Purpose: Displays interface status and IP addresses

7. Connectivity Testing

7.1 Intra-VLAN Testing

From PC0, ping PC1 (both in VLAN 10):

C:\> ping 192.168.10.3

Expected Result: Successful replies (devices in same VLAN communicate directly through switch)

7.2 Inter-VLAN Testing

From PC0 (VLAN 10), ping PC2 (VLAN 20):

C:\> ping 192.168.20.2

Expected Result: Successful replies (traffic routed through router subinterfaces)

7.3 Gateway Testing

From any PC, ping its default gateway:

From PC0: C:\> ping 192.168.10.1

From PC2: C:\> ping 192.168.20.1

Expected Result: Successful replies

8. Observations

• VLANs successfully created and ports assigned to respective VLANs.

• Trunk link established between switch and router for VLAN traffic.

• Devices within the same VLAN can communicate directly through the switch.

• Inter-VLAN communication requires routing through the router subinterfaces.

• Router-on-a-stick configuration successfully enables communication between VLANs.

• All connectivity tests passed without packet loss.

9. Result

VLANs were successfully configured on the switch with appropriate port assignments. Inter-VLAN routing was implemented using router subinterfaces with 802.1Q encapsulation. All devices can communicate within their VLANs and across VLANs through the router, demonstrating successful network segmentation and inter-VLAN routing.

10. Conclusion

This lab successfully demonstrated VLAN configuration, switchport mode configuration (access and trunk), and Inter-VLAN routing using the router-on-a-stick method. The lab illustrates how VLANs provide network segmentation for security and traffic management, while Inter-VLAN routing enables communication between different VLANs. Understanding these concepts is essential for designing scalable and secure enterprise networks.